Austin LaHue

Ph.D. Student, Human-Centered Computing
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Education

Clemson University, School of Computing — Clemson, SC
Doctor of Philosophy in Human-Centered Computing — May 2029 expected

- Advisor: Dr. Carlos Toxtli-Hernandez
- Research focus: Human-Centered AI, AI for mathematical discovery, AI-assisted journaling

The University of Texas at Austin, College of Natural Sciences — Austin, TX
Master of Science in Data Science — May 2024

Brigham Young University, Marriott School of Business — Provo, UT
Bachelor of Science in Global Supply Chain Management — June 2021

Research Interests

My research develops and evaluates human-centered AI systems that support reflection, reasoning, and creativity. I currently focus on AI-integrated systems for mathematical discovery, including conjecture generation and evaluation, and AI-assisted journaling, including personal and peer-based reflection platforms.

Research Experience

Human-AI Empowerment Lab, Clemson University, School of Computing — Clemson, SC
Graduate Research Assistant · Advisor: Dr. Carlos Toxtli-Hernandez — Aug. 2025–Present

- Designed and conducted a 15-day within-participant field study of AI-mediated peer journaling with 28 participants across private, manual-sharing, and AI-mediated conditions, collecting 218 journal entries and 120 peer responses.
- Developed an AI-mediated journaling platform with reviewable LLM mediation, privacy validation, peer exchange, and behavioral logging; across 72 AI-mediated sharing decisions, 95.8% of AI suggestions were accepted without editing, while mediation increased sharing-decision time by 2.82×.
- Developed an automated LLM pipeline for mathematical conjecture generation spanning 61 theorems from 12 research papers across four mathematical domains, producing and evaluating 793 conjectures under 13 experimental context conditions.
- Designed a paired ablation framework across six mathematical-context components to quantify how different forms of context affect LLM conjecture generation, revealing complementary roles for mathematical objects and structural constraints.

Brigham Young University, Marriott School of Business — Provo, UT
Research Assistant · Advisor: Dr. James Gaskin — Sep. 2022–Mar. 2024

- Proposed and designed an IRB-approved study, conducted literature review, constructed pre- and post-treatment survey instruments, and analyzed treatment/control data in Python.
- Identified a 15% increase in mental-health score in the treatment group and co-authored a complete research paper published in the ECIS 2024 proceedings.

Brigham Young University, Marriott School of Business — Provo, UT
Research Assistant · Advisor: Dr. Clark Pixton — June 2021

- Synthesized findings from 13 interviews on mask production and pop-up hospitals during the pandemic.
- Improved transcript readability by writing Python code to remove timestamps from interview transcripts.

Peer-Reviewed Publications

1. **LaHue, Austin** and Toxtli, Carlos. "SHARE: A Formative Probe of AI-Mediated Disclosure in Peer Journaling." *Proceedings of the 14th International Conference on Human-Agent Interaction (HAI '26)*, 2026. **In press.** doi:10.1145/3841580.3845678.
2. **LaHue, Austin**, Burr, Michael, Garcia Puente, Luis David, Jansari, Vinita Gangaram, Nye, Benjamin E., and Toxtli Hernandez, Carlos. "Which Mathematical Context Supports Target-Aligned Conjecture Generation? A Paired Ablation Study." *The Proceedings of the 4th Workshop on Mathematical Natural Language Processing (MathNLP 2026)*, co-located with EMNLP 2026, pp. 136–151. Budapest, Hungary: Association for Computational Linguistics, 2026. ACL Anthology.
3. Proctor, Harris, An, Li, **LaHue, Austin**, Burr, Michael, Jansari, Vinita Gangaram, Toxtli Hernandez, Carlos, Garcia Puente, Luis David, and Nye, Benjamin E. "NaturalPRISM: A Natural Language Premise Retrieval Task for Research-level Intermediate Proof States." *The Proceedings of the 4th Workshop on Mathematical Natural Language Processing (MathNLP 2026)*, co-located with EMNLP 2026, pp. 167–176. Budapest, Hungary: Association for Computational Linguistics, 2026. ACL Anthology.
4. **LaHue, Austin**, Gaskin, James, and Daniel, Kennedy. "Intermittent Digital Fasting: The Mental Health Benefits of a One-Hour Daily Social Media Hiatus." *ECIS 2024 Proceedings*, 10, 2024. AIS Electronic Library.

Presentations

- **LaHue, Austin.** "Which Mathematical Context Supports Target-Aligned Conjecture Generation? A Paired Ablation Study." Poster presentation, 4th Workshop on Mathematical Natural Language Processing (MathNLP 2026) at EMNLP 2026, Budapest, Hungary, Oct. 28, 2026.
- **LaHue, Austin.** "Which Mathematical Context Supports Target-Aligned Conjecture Generation? A Paired Ablation Study." Poster presentation, HPC Day 2026, Clemson University, Clemson, SC, Sep. 18, 2026.
- Nye, B., **LaHue, A.**, Burr, M. "AIMing: Interactive Conjecture Proving." NSF AIMing Principal Investigator Meeting, Carnegie Mellon University, Pittsburgh, PA, Apr. 2, 2026.
- **LaHue, Austin.** Poster presentation, INTEGRATE Symposium Poster Session, University of Wisconsin–Madison, Madison, WI, Mar. 24, 2026.
- **LaHue, Austin.** "Live Demo: Visualizing Survey Data in Tableau." Disney Tableau Day, Orlando, FL, June 2024.

Teaching Experience and Interests

Brigham Young University, Marriott School of Business Provo, UT
Teaching Assistant · GSCM 412: Operations Analytics Dec. 2020–Apr. 2021

- Supported students in exploring, visualizing, and interpreting 2 million rows of data using Tableau, SQL, and Solver.
- Reviewed 315 assignments and provided feedback on missed questions, analytical interpretation, and dashboard effectiveness.

Teaching interests: Human-Centered AI; Human-Computer Interaction; HCC Research Methods; Data Visualization; Applied Data Science.

Funding and Research Events

- Graduate Research Assistant, 20 hours/week, Dr. Carlos Toxtli-Hernandez, Clemson University, Fall 2025–Spring 2026; Fall 2026.
- Graduate Student Summer Employee, 20 hours/week, Dr. Carlos Toxtli-Hernandez, Clemson University, Summer 2026.

- NSF AIMing Principal Investigator Meeting, Carnegie Mellon University, Apr. 2–3, 2026
- NSF Research Traineeship (NRT+) Summit, University of Wisconsin–Madison, Mar. 23–25, 2026

Professional Service and Leadership

- Advisor, Trust and Safety Institute, June 2026–Present. Provide advisory input on trust and safety, responsible AI, user empowerment, platform governance, and digital safety from a human-centered AI research perspective.
- Treasurer, School of Computing Graduate Student Association, Clemson University, 2026–Present.
- Peer Reviewer, PACIS 2024, Track 12: Human-Centric Information Systems Design, Development, and Use; HAI 2026 Poster Session.
- Volunteer Representative, The Church of Jesus Christ of Latter-day Saints, Durban, ZA, 2016–2018.

Honors and Awards

- 1st Place, Dataiku Everyday AI Data Visualization Challenge, 2025.
- Winner, Disney Tableau Day Viz Challenge, 2024.

Technical Skills

- **AI/ML and NLP:** large language models, agentic AI, prompt/evaluation pipelines, LoRA, supervised fine-tuning, PyTorch, XGBoost, natural language inference, predictive modeling.
- **Programming and Data Systems:** Python, SQL, JavaScript, Snowflake, Dataiku, data pipelines, statistical analysis, Git/GitHub.
- **Human-Centered Research:** human-subjects research, IRB protocols, survey design, experimental design, pre/post intervention studies, measurement and evaluation of HCC systems, literature synthesis.
- **Analytics and Visualization:** Tableau, dashboard development, data visualization, audience analytics, storytelling with data, marketing analytics.
- **Certifications:** Dataiku Core Designer (2022), ML Practitioner (2022), Advanced Designer (2024).

Professional Experience

The Walt Disney Company Orlando, FL
Business Intelligence Analyst / Data Scientist Jan. 2022–Aug. 2025

- Predicted annual passholder purchase likelihood using an XGBoost model to support Walt Disney World marketing campaigns.
- Managed development of 14 RFM models to enhance direct-channel reach across Disney franchises.
- Raised email engagement by 42% by retraining a predictive model for guest likelihood to purchase an Alaska cruise.
- Developed dashboards summarizing opt-in preferences for 110 million Disney guests and supporting audience planning with ticket usage, resort stays, demographics, and contactability.
- Built SQL, Snowflake, Dataiku, and Tableau data pipelines and analyses for strategy partners, including quantifying the effect of 18 email suppressions on the WDW Annual Passholder contactable database.

The Church of Jesus Christ of Latter-day Saints Salt Lake City, UT
Strategy Analyst July 2021–Dec. 2021

- Led five strategic initiatives, analyzed 34,000+ work-order records in Tableau to identify cost-reduction opportunities, and visualized 1,400+ employee-satisfaction survey responses for 14 international areas.

Walmart eCommerce Davenport, FL
Supply Chain Intern May 2020–Aug. 2020

- Reduced order-filling rework by 75% by implementing multi-face slots to reduce item shortages and conducted trend analysis of yearly rework for four high-demand items.